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Litterature:

H. H. Hamedani, *Probability and Statistics: The Science of Uncertainty*, 3rd ed.,

<https://www.probabilitycourse.com>

Sampling Distribution

Problem with point estimates

Point estimates has variance –
in repeated experiments the estimate change

Repeated point estimates

A possible solution: Collect many Point estimates and calculate the the resulting distribution

Why we *don't* do that in practice

In real life, we almost never have the luxury of taking many independent samples.

Usually we only have **one dataset** — one random sample.

Solution:

Use probability theory to **approximate** what that spread *would* look like, and then express it as an **interval** around our one observed estimate.

Interval Estimation – Confidence vs Precision

- Example of a point estimate:

$$\hat{\theta} = 34.25$$

- Example of an interval estimate:

$$[\hat{\theta}_\ell, \hat{\theta}_h] = [30.69, 37.81]$$

Here, $\hat{\theta}_\ell$ is the **lower bound** and $\hat{\theta}_h$ is the **upper bound** of the interval. We hope the true θ lies somewhere in this range.

Two Key Features of Interval Estimation

1. Length of the interval

$$\hat{\theta}_h - \hat{\theta}_\ell$$

This measures the **precision** of our estimate.

- A shorter interval = more precise estimate.
- A longer interval = less precise.

2. Confidence level

Higher confidence levels are desirable, but they usually come at the cost of **longer intervals**.

Interval Estimation – the Idea

Setup

Let

$$X_1, X_2, \dots, X_n$$

be a random sample from a distribution with an unknown parameter θ . Our goal is to construct **two estimators** for θ :

1. **Lower estimator:**

$$\hat{\theta}_\ell = \hat{\theta}_\ell(X_1, X_2, \dots, X_n),$$

2. **Upper estimator:**

$$\hat{\theta}_h = \hat{\theta}_h(X_1, X_2, \dots, X_n).$$

Together, they form the **interval estimator**

$$[\hat{\theta}_\ell, \hat{\theta}_h].$$

Confidence Level

The estimators $\hat{\theta}_\ell$ and $\hat{\theta}_h$ are chosen so that the probability that the interval contains the true parameter θ is at least:

$$P\left(\hat{\theta}_\ell \leq \theta \leq \hat{\theta}_h\right) \geq 1 - \alpha.$$

- Here, $1 - \alpha$ is called the **confidence level**.
- Common choices:
 - $\alpha = 0.10 \Rightarrow 90\%$ confidence
 - $\alpha = 0.05 \Rightarrow 95\%$ confidence
 - $\alpha = 0.01 \Rightarrow 99\%$ confidence

Interval Estimation – the Idea

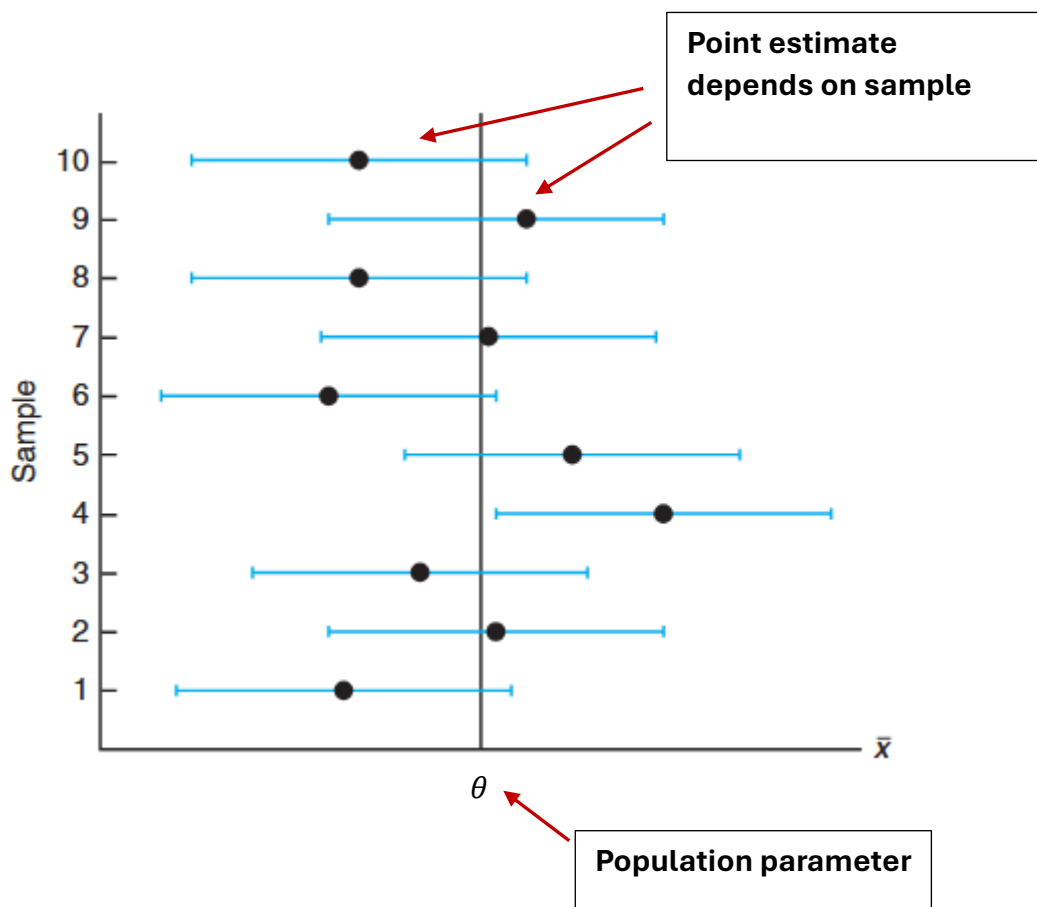
Example:

To find a 95% confidence interval for θ , we need to choose $\hat{\theta}_\ell$ and $\hat{\theta}_h$ such that

$$P\left(\hat{\theta}_\ell \leq \theta \leq \hat{\theta}_h\right) \geq 0.95.$$

Interpretation

- The **randomness** comes from the estimators $\hat{\theta}_\ell$ and $\hat{\theta}_h$, since they are based on the random sample $(X_1, \dots, X_n)$.
- The parameter θ itself is **fixed but unknown**.
- Different samples will produce different confidence intervals, but in the long run, about a proportion $1 - \alpha$ of those intervals will contain the true value of θ .



Interval Estimation – Finding Bounds

Finding Interval Estimators

Starting Point

To construct confidence intervals, we need a way to find bounds (x_ℓ, x_h) such that a random variable X falls between them with high probability.

For a continuous random variable X with CDF $F_X(x)$:

$$P(x_\ell \leq X \leq x_h) = F_X(x_h) - F_X(x_\ell) = 1 - \alpha$$

Symmetric Choice of Bounds

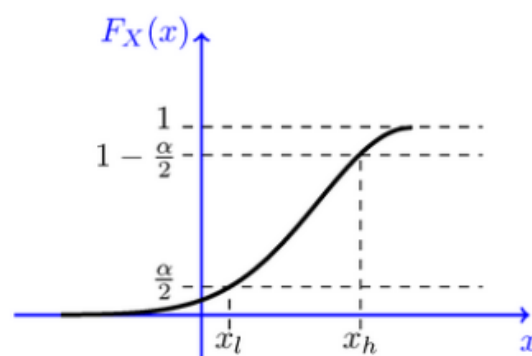
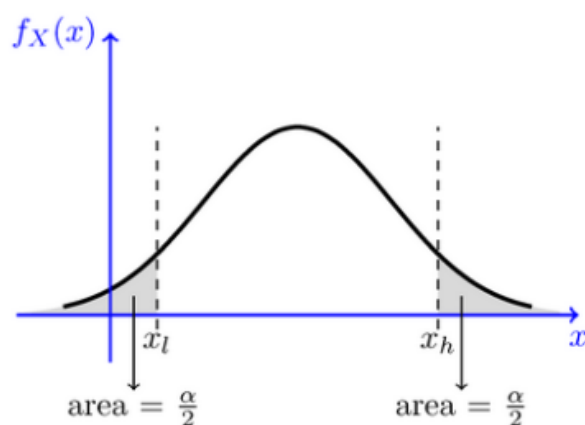
A common choice is:

$$F_X(x_\ell) = \frac{\alpha}{2}, \quad F_X(x_h) = 1 - \frac{\alpha}{2}.$$

So:

$$x_\ell = F_X^{-1}\left(\frac{\alpha}{2}\right), \quad x_h = F_X^{-1}\left(1 - \frac{\alpha}{2}\right).$$

This interval $[x_\ell, x_h]$ is a $(1 - \alpha)$ interval for X .



Interval Estimation – Finding Bounds

Example: 95% Confidence interval for Standard Normal

Find x_ℓ, x_h such that:

$$P(x_\ell \leq Z \leq x_h) = 0.95, \quad Z \sim N(0, 1).$$

Solution:

- $\alpha = 0.05$.
- $x_\ell = \Phi^{-1}(0.025) = -1.96$.
- $x_h = \Phi^{-1}(0.975) = 1.96$.

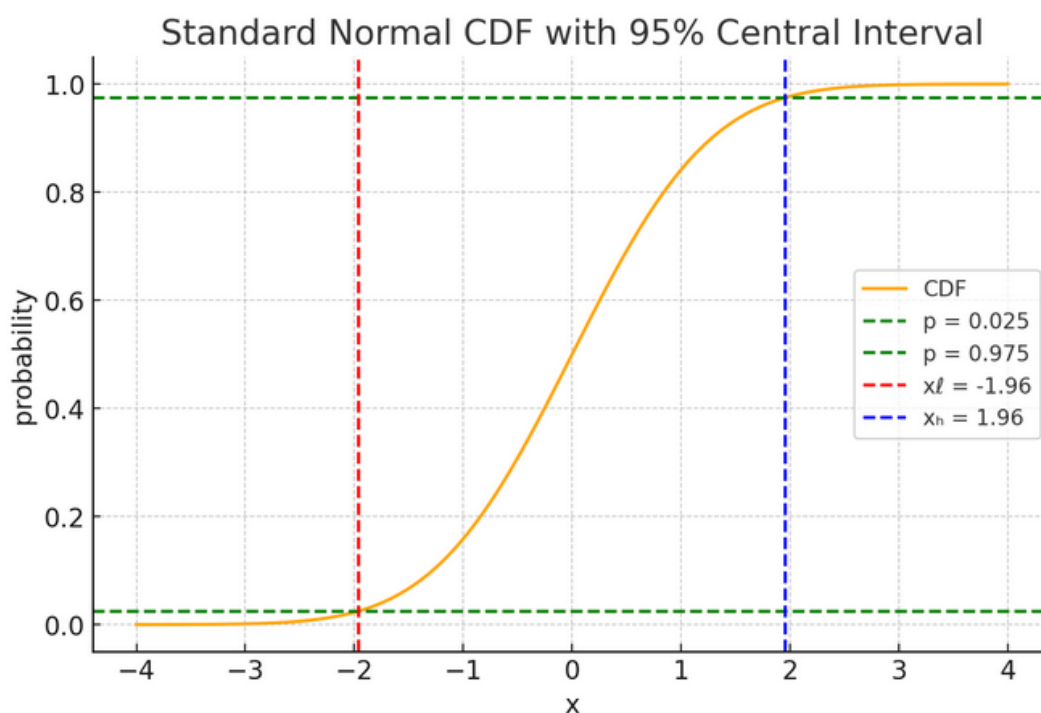
Thus:

$$P(-1.96 \leq Z \leq 1.96) = 0.95.$$

python

```
from scipy.stats import norm

# Lower and upper quantiles for 95% CI
x_l = norm.ppf(0.025) # \Phi^{-1}(0.025)
x_h = norm.ppf(0.975) # \Phi^{-1}(0.975)
```



Interval Estimation – known variance

Example: 95% Confidence interval for Large n, Known Variance

Let

$$X_1, X_2, \dots, X_n \text{ i.i.d. with } E[X_i] = \mu = \theta, \quad \text{Var}(X_i) = \sigma^2.$$

We want to find a 95% confidence interval for θ .

Solution:

- From the Central Limit Theorem, the standardized sample mean

$$Z_n = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

approximately follows a standard normal distribution:

$$Z_n \sim N(0, 1) \quad \text{for large } n.$$

- For a 95% confidence level:

$$P(-1.96 \leq Z_n \leq 1.96) \approx 0.95.$$

- Substituting Z_n :

$$P\left(-1.96 \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq 1.96\right) \approx 0.95.$$

- Rearranging for $\mu = \theta$:

$$P\left(\bar{X} - \frac{1.96\sigma}{\sqrt{n}} \leq \theta \leq \bar{X} + \frac{1.96\sigma}{\sqrt{n}}\right) \approx 0.95.$$

Final Confidence Interval:

$$\left[\bar{X} - \frac{1.96\sigma}{\sqrt{n}}, \bar{X} + \frac{1.96\sigma}{\sqrt{n}} \right].$$

Interval Estimation – Known Variance

Example: 99% Confidence interval for Large n, Known Variance

- For a 99% confidence level, $\alpha = 0.01$ and $z_{\alpha/2} = z_{0.005} = 2.576$:

$$P(-2.576 \leq Z_n \leq 2.576) \approx 0.99.$$

- Substitute Z_n and solve for $\mu = \theta$:

$$P\left(-2.576 \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq 2.576\right) \approx 0.99$$

$$\Rightarrow P\left(\bar{X} - \frac{2.576 \sigma}{\sqrt{n}} \leq \theta \leq \bar{X} + \frac{2.576 \sigma}{\sqrt{n}}\right) \approx 0.99.$$

Final Confidence Interval:

$$\left[\bar{X} - \frac{2.576 \sigma}{\sqrt{n}}, \bar{X} + \frac{2.576 \sigma}{\sqrt{n}} \right]$$

Quiz

We have i.i.d. samples $X_1, X_2, \dots, X_n$ with

$$E[X_i] = \mu = \theta, \quad \text{Var}(X_i) = \sigma^2.$$

Question:

Which of the following is the correct **90% confidence interval** for θ ?

A)

$$\left[\bar{X} - \frac{1.645\sigma}{\sqrt{n}}, \bar{X} + \frac{1.645\sigma}{\sqrt{n}} \right]$$

B)

$$\left[\bar{X} - \frac{1.96\sigma}{\sqrt{n}}, \bar{X} + \frac{1.96\sigma}{\sqrt{n}} \right]$$

C)

$$\left[\bar{X} - \frac{2.326\sigma}{\sqrt{n}}, \bar{X} + \frac{2.326\sigma}{\sqrt{n}} \right]$$

D)

$$\left[\bar{X} - \frac{1.28\sigma}{\sqrt{n}}, \bar{X} + \frac{1.28\sigma}{\sqrt{n}} \right]$$

Interval Estimation – Known Variance**General Confidence Interval Formula (Large n , Known Variance)**

For i.i.d. samples $X_1, X_2, \dots, X_n$ with

$$E[X_i] = \mu = \theta, \quad \text{Var}(X_i) = \sigma^2,$$

a $(1 - \alpha)100\%$ confidence interval for θ is given by

$$\left[\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right].$$

When to use:

- Data are i.i.d and sample size n is large $\Rightarrow$ CLT applies
- The population variance σ^2 is known.

Interval Estimation – Known Variance**Example: Determining Sample Size**

Let $X_1, X_2, \dots, X_n$ be i.i.d. measurements modeled as

$$X_i = \theta + W_i,$$

where

$$E[W_i] = 0, \quad \text{Var}(W_i) = 4.$$

Thus

$$E[X_i] = \theta, \quad \text{Var}(X_i) = 4.$$

We estimate θ using the sample mean

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

Required Sample Size

We want the estimation error to be at most **0.25 with 90% confidence**.

For known variance, the 90% confidence interval is

$$\bar{X} \pm z_{0.05} \frac{\sigma}{\sqrt{n}}.$$

The **margin of error** is therefore

$$z_{0.05} \frac{\sigma}{\sqrt{n}}.$$

We require

$$z_{0.05} \frac{\sigma}{\sqrt{n}} = 0.25.$$

Given

$$\sigma = 2, \quad z_{0.05} = 1.645,$$

we obtain

$$1.645 \cdot \frac{2}{\sqrt{n}} = 0.25.$$

Solving:

$$\sqrt{n} = \frac{1.645 \cdot 2}{0.25} = 13.16$$

$$n \approx 173.3.$$

Therefore,

$$n \geq 174.$$

Quiz

Previously we determined that to achieve a **margin of error 0.25** with **90% confidence**, the required sample size is

$$n \geq 174.$$

Now suppose we repeat the calculation with **95% confidence** (all other quantities unchanged).

Question:

What happens to the required sample size n ?

- A) n stays the same
- B) n decreases
- C) n increases
- D) It is impossible to determine without knowing θ

Interval Estimation – Unknown Variance

We now **estimate** σ from the data using sample **variance**.

The **sample variance** is defined as:

$$S^2 = \frac{1}{n-1} \sum_{k=1}^n (X_k - \bar{X})^2.$$

Thus, the **sample standard deviation** is:

$$S = \sqrt{S^2} = \sqrt{\frac{1}{n-1} \sum_{k=1}^n (X_k - \bar{X})^2}.$$

For large n , S is a good estimate of σ .

Approximate confidence interval:

$$\left[\bar{X} - z_{\alpha/2} \frac{S}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{S}{\sqrt{n}} \right]$$

When to use:

- Data are i.i.d and sample size **n is large** $\Rightarrow$ CLT applies
- The population variance **σ^2 is unknown**.

Degree of Freedom (df)

When we compute the sample variance

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2,$$

we divide by $n - 1$ because **one piece of information** has already been used to compute the mean.

So we say sample mean has **$n-1$ degrees of freedom**

Contrast: when the true mean is known

Suppose the population mean μ is known.

Then we could estimate the variance using

$$\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2.$$

Now the deviations satisfy **no constraint** like before.

Each of the n terms

$$(X_1 - \mu), \dots, (X_n - \mu)$$

can vary freely.

So the variance estimate has **n degrees of freedom**.

T-distribution

The CLT works when the sample size **n is large**, since it tells us that the sample mean is **approximately normal**:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

However, what if:

- n is **small**, and
- the population variance σ^2 is **unknown**?

In this case, the CLT approximation is no longer sufficient, and the substitution of S for σ changes the distribution.

This leads to the **Student's t-distribution**:

If

$$X_1, X_2, \dots, X_n \sim N(\mu, \sigma^2),$$

and

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2,$$

then the statistic

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

follows a **t-distribution** with $n - 1$ **degrees of freedom**:

$$T \sim t(n - 1)$$

T-distribution

Properties of the t-Distribution

1. Shape:

The t-distribution is **symmetric** and **bell-shaped**, centered at 0.

2. Mean:

$$E[T] = 0, \quad \text{for } n > 1$$

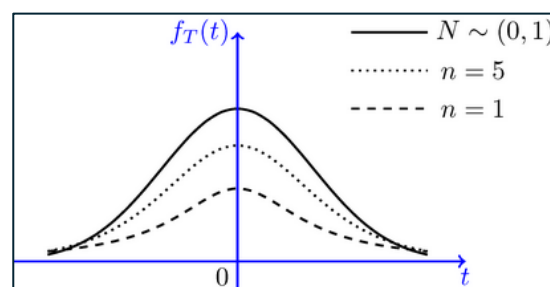
3. Variance:

$$\text{Var}(T) = \frac{n}{n-2}, \quad \text{for } n > 2$$

4. Convergence to Normal:

As the sample size increases, the t-distribution becomes indistinguishable from the standard normal:

$$t(n) \xrightarrow{n \rightarrow \infty} N(0, 1)$$



5. For any $p \in [0, 1]$ and $n \in \mathbb{N}$, we define $t_{p,n}$ as the real value for which

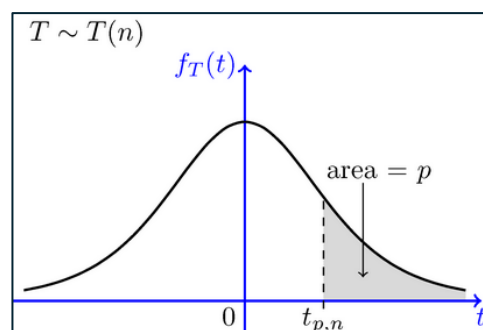
$$P(T > t_{p,n}) = p.$$

Since the t-distribution has a symmetric PDF, we have

$$t_{1-p,n} = -t_{p,n}.$$

python

```
from scipy.stats import t
t_crit = t.ppf(1 - alpha/2, df)
```



Interval Estimation – Unknown Variance and small n

Confidence interval for T-distribution:

$$P\left(-t_{\frac{\alpha}{2}, n-1} \leq T \leq t_{\frac{\alpha}{2}, n-1}\right) = 1 - \alpha.$$

Therefore,

$$P\left(-t_{\frac{\alpha}{2}, n-1} \leq \frac{\bar{X} - \mu}{S/\sqrt{n}} \leq t_{\frac{\alpha}{2}, n-1}\right) = 1 - \alpha,$$

which is equivalent to

$$P\left(\bar{X} - t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}} \leq \mu \leq \bar{X} + t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}}\right) = 1 - \alpha.$$

We conclude that

$$\left[\bar{X} - t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}}, \bar{X} + t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}}\right]$$

is a $(1 - \alpha)100\%$ confidence interval for μ .

When to use:

- The population standard deviation **σ is unknown**.
- The sample size is **small** (typically $n < 30$).
- The sample comes from a **normally distributed population**.

Interval Estimation – Unknown Variance and small n

Example

A farmer weighs 10 randomly chosen watermelons (lbs):

7.72, 9.58, 12.38, 7.77, 11.27, 8.80, 11.10, 7.80, 10.17, 6.00

Assuming weights are **normally distributed** with mean μ and unknown variance σ^2 , find a **95% confidence interval** for μ .

From the data:

$$\bar{X} = 9.259 \approx 9.26, \quad S^2 = 3.961 \approx 3.96, \quad n = 10, \quad \alpha = 0.05.$$

Degrees of freedom = $n - 1 = 9$.

Critical value:

$$t_{0.025,9} = \text{t.ppf}(0.975, 9) \approx 2.262.$$

The 95% CI is

$$\begin{aligned} \left[\bar{X} - t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}}, \bar{X} + t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}} \right] &= \left[9.259 - 2.262 \cdot \frac{\sqrt{3.961}}{\sqrt{10}}, 9.259 + 2.262 \cdot \frac{\sqrt{3.961}}{\sqrt{10}} \right] \\ &= [7.84, 10.68] \end{aligned}$$

Interval Estimation – Unknown Variance and small n

Example

Same example as the above fully solved in Python:

```
import numpy as np
from scipy.stats import t

# Sample data (weights in lbs)
x = np.array([7.72, 9.58, 12.38, 7.77, 11.27, 8.80, 11.10, 7.80, 10.17, 6.00])

# Sample statistics
n = len(x)                # sample size
x_bar = np.mean(x)        # sample mean
s2 = np.var(x, ddof=1)    # sample variance (ddof=1 → divide by n-1)
s = np.sqrt(s2)           # sample standard deviation

# Confidence Level
alpha = 0.05
df = n - 1                # degrees of freedom

# Critical t-value (two-tailed)
t_crit = t.ppf(1 - alpha/2, df)

# Confidence interval
margin = t_crit * s / np.sqrt(n)
ci_lower = x_bar - margin
ci_upper = x_bar + margin
```

Quiz

If the sample size n is small, using the z -value (from `scipy.stats.norm.ppf`) instead of the t -value will typically make the interval:

- A: Wider than it should be.
- B: Narrower than it should be.
- C: Shifted upward.
- D: Symmetric but centered at zero.
- E: Don't know

Quiz

In Python, if we wanted to correctly handle a small sample (say $n = 12$), which line should we change in the code?

- A. Replace `np.mean(x)` with `np.median(x)` .
- B. Replace `norm.ppf(1 - alpha / 2)` with `t.ppf(1 - alpha / 2, df=n-1)` .
- C. Replace `np.std(x, ddof=1)` with `np.std(x)` .
- D. Replace `np.sqrt(n)` with `np.sqrt(n-1)` .
- E. Don't know

From Probability to Inference

Example:

In a communication system, each data packet consists of 1000 bits. Each bit is received in error with probability $p = 0.1$, and bit errors occur independently.

We receive one packet and observe **120 bit errors**.

Question:

Does the observed value 120 fall within the **central 95% of the distribution of X** under the model $X \sim \text{Binomial}(1000, 0.1)$?

Each bit error $X_i \sim \text{Bernoulli}(p = 0.1)$.

The total number of bit errors:

$$Y = \sum_{i=1}^{1000} X_i \sim \text{Binomial}(n = 1000, p = 0.1).$$

So,

$$E[Y] = np = 100, \quad \text{Var}(Y) = np(1 - p) = 90.$$

By the **Central Limit Theorem**, for large n :

$$\frac{Y - np}{\sqrt{np(1 - p)}} \approx N(0, 1).$$

Thus, a 95% confidence interval for the expected number of errors is

$$np \pm 1.96\sqrt{np(1 - p)}.$$

Compute the interval:

$$100 \pm 1.96\sqrt{90} = 100 \pm 1.96 \times 9.487 = 100 \pm 18.59.$$

$$\boxed{(81.4, 118.6)}.$$

From Probability to Inference

Result:

$120 > 118.6$, so it lies **just outside** the 95% confidence interval.

Interpretation

At the 95% confidence level, 120 errors would be considered **unusually high** if the true bit-error probability were $p = 0.1$.